

Daily flaxseed consumption improves glycemic control in obese men and women with pre-diabetes: a randomized study.

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The study hypothesis was that fasting glucose, insulin, fructosamine, C-reactive protein, and interleukin-6 decrease and adiponectin increases with daily flaxseed consumption in overweight or obese individuals with pre-diabetes. In this randomized, cross-over study overweight or obese men and postmenopausal women (n = 25) with pre-diabetes consumed 0, 13, or 26 g ground flaxseed for 12 weeks. Glucose, insulin, homeostatic model assessment (HOMA-IR), and normalized percent of α -linolenic fatty acid (ALA) were significantly different by treatment (multiple analysis of variance, $P = .036$, $P = .013$, $P = .008$, $P = .024$ respectively). Paired t tests showed glucose decreased on the 13 g intervention compared to the 0 g period [13 g = -2.10 ± 1.66 mg/L (mean $\pm$ SEM), 0 g = 9.22 ± 4.44 mg/L, $P = .036$]. Insulin decreased on the 13 g intervention but not the 26 g ($P = .021$) and 0 g ($P = .013$) periods (13 g = -2.12 ± 1.00 mU/L, 26 g = 0.67 ± 0.84 mU/L, 0 g = 1.20 ± 1.16 mU/L). HOMA-IR decreased on the 13 g period but not on the 26 g ($P = .012$) and 0 g ($P = .008$) periods (13 g = -0.71 ± 0.31 , 26 g = 0.27 ± 0.24 , 0 g = 0.51 ± 0.35). The α -linolenic fatty acid decrease for the 0 g period was different than the 13 g ($P = .024$) and 26 g ($P = .000$) periods (13 g = 0.20 ± 0.04 , 26 g = 0.35 ± 0.07 , 0 g = -0.01 ± 0.07). Fructosamine, high sensitivity C-reactive protein, adiponectin, and high-sensitivity interleukin-6 had no significant differences. Flaxseed intake decreased glucose and insulin and improved insulin sensitivity as part of a habitual diet in overweight or obese individuals with pre-diabetes. Copyright © 2013 Elsevier Inc. All rights reserved.